

Multi-Page Navigation - Code Logic

Task: Create a 3-page Flutter flow consisting of a Splash Screen, Login Page, and Dashboard. Navigation must move between the screens and the username entered on Login must be passed to Dashboard.

1. Splash Screen

App entry point

Wait briefly, then open Login



2. Login Page

Enter username

Validate Login button

↓ Pass username String

3. Dashboard

Receive username in constructor

Display Welcome, username

1. main.dart - Application Entry

main.dart starts the Flutter application and sets SplashScreen as the first page. This means the user cannot jump directly to Login or Dashboard when the app launches.

```
MaterialApp(  
  debugShowCheckedModeBanner: false,  
  home: const SplashScreen(),  
)
```

2. splash.dart - Automatic Navigation

SplashScreen is Stateful because it performs an action after the widget is created. initState() starts a short delay, then Navigator.pushReplacement() opens LoginPage. pushReplacement is used so pressing Back from Login does not return to the splash screen.

```
@override  
void initState() {  
  super.initState();  
  
  Future.delayed(const Duration(seconds: 2), () {  
    if (!mounted) return;  
  
    Navigator.pushReplacement(  
      context,  
      MaterialPageRoute(  
        builder: (context) => const LoginPage(),  
      ),  
    );  
  });  
}
```

3. login.dart - Input, Validation, and Data Passing

LoginPage uses a TextEditingController to read the username from a TextField. When Login is pressed, empty input produces a Snackbar. Valid input creates DashboardPage and passes the username through its required constructor parameter.

```
final username = usernameController.text.trim();  
  
if (username.isEmpty) {  
  ScaffoldMessenger.of(context).showSnackBar(  
    const SnackBar(  
      content: Text('Please enter your username.'),  
    ),  
  );  
  return;  
}  
  
Navigator.pushReplacement(  
  context,  
  MaterialPageRoute(  
    builder: (context) => DashboardPage(  
      username: username,  
    ),  
  ),  
);
```

4. dashboard.dart - Receiving the Username

DashboardPage declares a final String username. The constructor requires that value. Flutter therefore receives the exact username supplied by LoginPage and can display it in the UI.

```
class DashboardPage extends StatelessWidget {  
  final String username;  
  
  const DashboardPage({  
    super.key,  
    required this.username,  
  });  
  
  // ...  
  Text('Welcome, $username')  
}
```

Navigation Logic Summary

Step	Source	Action	Destination / Result
1	main.dart	Launch app	SplashScreen
2	SplashScreen	2-second delay	LoginPage
3	LoginPage	Enter username	String stored from TextField
4	Login button	Validate + navigate	DashboardPage(username: username)
5	Dashboard	Read constructor value	Shows "Welcome, <username>"

Expected Test

Launch the app. The splash screen should automatically open Login. Enter a username such as "Shayma" and press Login. Dashboard should open and display "Welcome, Shayma". Entering an empty username should keep the user on Login and show a Snackbar.